

How Strong Is India's Military Industry? Actually Not as Weak as People Think

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Compared with China's, India's military industry is certainly far behind. But its level is much stronger than many people on Zhihu imagine. China and India have territorial disputes, and may also compete industrially in the future; so it is necessary for Chinese people to understand India's military strength, to avoid miscalculation — and never to speak lightly of war.

Below is a brief introduction.

Missiles

Hypersonic Missiles

On 17 November 2024, India's Defence Research and Development Organisation (DRDO) successfully conducted the first flight test of a long-range hypersonic missile. Launched from Dr. Abdul Kalam Island off the coast of Odisha, the missile can carry various payloads with a range of over 1,500 kilometres, an important addition to India's defence arsenal. Existing missile-defence systems find hypersonic missiles very hard to stop. From northeastern India, 1,500 kilometres already covers the major cities of China's southwest. If India later develops hypersonic missiles of two or three thousand kilometres or more, the threat to China will be even greater.

Hypersonic Missile

Can reach speed in excess of Mach 5,
has range of over 1,500 km

Unlike traditional ballistic &
cruise missiles, hypersonic
weapon almost impossible to
intercept by current air defence
systems

To be used
in non-nuclear
role, expected to
have anti-ship &
land attack
variants

New weapon
developed for
upcoming tri
services
rocket force



India joins
elite group of
nations with
such capacity,
other
countries
include US,
Russia and
China



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Some say this Indian hypersonic missile is a fake, because India has no wind tunnel. But India does have wind tunnels. The wind tunnel developed by the IITs reaches Mach 8.8 to Mach 29.4; even taking the low value of Mach 8.8, it is equivalent to China's JF-10 hypersonic shock wind tunnel, completed in 1998 and put into operation that year. India's wind tunnel is enough to support hypersonic missile development.

IIT Kanpur achieves major milestone with India's First Hypervelocity Expansion Tunnel Test Facility

Kanpur, 05 February, 2024: The Indian Institute of Technology Kanpur (IITK) has successfully established and tested India's first Hypervelocity Expansion Tunnel Test Facility, a major achievement that puts India amongst only a handful of countries with this advanced hypersonic testing capability. The facility, named S2, is capable of generating flight speeds between 3-10 km/s, simulating the hypersonic conditions encountered during atmospheric entry of vehicles, asteroid entry, scramjet flights, and ballistic missiles. This makes it a valuable test facility for ongoing missions of ISRO and DRDO including Gaganyaan, RLV, and hypersonic cruise missiles.

The S2, nicknamed 'Jigarthanda', is a 24-meter-long facility located at IIT Kanpur's Hypersonic Experimental Aerodynamics Laboratory (HEAL) within the Department of Aerospace Engineering. The S2 was indigenously designed and developed over a period of three years with funding and support from the Aeronautical Research and Development Board (ARDB), the Department of Science and Technology (DST), and IIT Kanpur.

Commenting on this, **Prof. S. Ganesh, Director, IIT Kanpur**, said, "The successful establishment of S2, India's first hypervelocity expansion tunnel test facility, marks a historic milestone for IIT Kanpur and for India's scientific capabilities. I congratulate Prof. Sugarno and his team for their exemplary work in designing and fabricating the hypersonic research infrastructure. S2 will empower India's space and defence organizations with domestic hypersonic testing capabilities for critical projects and missions."

Prof. Mohammed Ibrahim Sugarno, Associate Professor, Department of Aerospace Engineering and Centre for Lasers & Photonics at IIT Kanpur said, "Building S2 has been extremely challenging, requiring in-depth knowledge of physics and precision engineering. The most crucial and challenging aspect was perfecting the 'free piston driver' system, which requires firing a piston at high pressure between 20-35 atmospheres down a 6.5 m. compression tube at speeds of 150-200 m/s, and bringing it to a complete stop or 'soft landing' at the end."

"However, with our expertise, we were able to overcome this. Our team is proud to have designed, built, and tested this one-of-a-kind facility, cementing India's position in the elite global hypersonic research community," added.

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Ballistic Missiles

That India has ballistic missiles is known to everyone, but assessments of their performance tend to be negative. In fact, India's ballistic missiles are developing fast and already have road-mobile and rail-mobile launch capability. This means India's second-strike capability is greatly enhanced: in a possible conflict, missile systems hidden in eastern India could be launched on the move, inflicting major damage on Chinese western cities such as Chengdu and Kunming. This year, the Agni-5 with a range of 5,000 kilometres was successfully test-fired; from Indian soil it can directly cover most Chinese cities. Even with China's strong interception capability, a single city hit by even a few missiles would suffer great casualties.

9月24日，印度国防研究与发展组织（DRDO）宣布，铁路机动发射系统成功试射一枚“烈火-P（Agni-Prime）”中程弹道导弹。



▲ 印度“烈火-P”中程弹道导弹试射，导弹飞出发射筒的瞬间

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The Agni-P missile has a design range of 1,000–2,000 km. According to DRDO, the road-mobile variant has been fielded with the Indian military. As planned, the Agni-P will replace India's earlier Agni-I (700 km) and Agni-II (2,000 km) missiles, or supplement them. According to the British think tank IISS, India currently fields 12 Agni-I launchers and 8 Agni-II launchers. As part of India's nuclear forces, the Agni-P has great significance as a rail-mobile deployment force. After all, a rail-based ballistic missile launch system provides India with a relatively cheap way of deploying ballistic missiles, greatly reducing their vulnerability to pre-emptive strikes or retaliation. Operationally, the rail-mobile system will make full use of India's extremely extensive rail network — about 64,000 km in total — enabling missiles to disperse quickly, making them hard for any adversary to detect and monitor. Railway tunnels are scattered throughout India, providing ready-made hardened shelters for the rail-mobile system, convenient for concealment. This not only makes them harder for the enemy to destroy, but also hard to track. There are signs that this Agni-P rail-mobile test drew particular Chinese attention: China's space-tracking ship Yuanwang-5 appeared in the Indian Ocean before the launch. Source: "India's Agni-P rail-mobile system test-fired; our Yuanwang-5 space-tracking ship monitors" (WeChat article)

According to Army Recognition on 21 August, based on information released by India's defence ministry, the successful flight test of the Agni-5 intermediate-range ballistic missile has been

confirmed. The launch was conducted at the Integrated Test Range in Chandipur, Odisha, under the Strategic Forces Command, which oversees India's nuclear operations. This user-training launch met all technical parameters, demonstrating combat readiness under realistic operational conditions and strengthening India's capability for long-range precision strikes as part of its nuclear deterrence posture. The Agni-5 is the most advanced missile in India's Agni series, representing a key technological leap in the country's ballistic missile programme. The missile is about 17.5 m long, about 2 m in diameter, and weighs about 50,000 kg in total. It uses a three-stage solid-propellant propulsion system, significantly improving its range, acceleration, and reliability over earlier systems. The missile is road-mobile, housed in a sealed launch canister mounted on a high-mobility launch vehicle, capable of storage, rapid deployment, and cold launch from varied terrain. It is equipped with a ring laser gyroscope-based inertial navigation system (RLG-INS), aided by India's regional navigation satellite system (IRNSS), ensuring a circular error probable (CEP) of 10–20 metres. The re-entry vehicle uses carbon-carbon composite shielding, allowing it to withstand temperatures above 3,000°C on terminal re-entry. Its range is estimated at about 5,800 km, capable of striking the entire Asian continent.

Source: "India successfully launches Agni-5 missile covering all of Asia" (WeChat article)

India's K-series submarine-launched ballistic missiles are currently too short in range, mainly for use against Pakistan. But the K-5 under development has a range of 5,000 kilometres; launched from the deep waters of the Indian Ocean, it could directly strike China's major cities, very hard to defend against. India is also developing the K-6, with an estimated range of 8,000 kilometres; if fielded, the threat would be even greater.

印度成功测试K-5潜射导弹的第二级火箭发动机

Original 兵器知识杂志社 兵器知识杂志 2025年12月3日 17:30 北京

印度的纳西克能源材料先进中心 (ACEM^Q) 宣布，他们已成功完成K-5潜射弹道导弹第二级火箭发动机的静态测试，验证了这种10吨级推力发动机的弹道性能、安全性、子系统集成，核心技术指标符合预期。



该发动机长2.68米，直径2.49米。它以T700碳纤维、Epofine树脂复合材料^Q制作外壳，内装HD-1.3复合推进剂。测试中采用热源型点火系统，喷口实现了 $\pm 2^\circ$ 的转向，250余组仪器数据与预测性能曲线高度吻合。预计在2026年中，将启动K-5导弹的水下发射试验。

K-5潜射导弹最大射程达5000~8000千米，战斗部重2000千克，最大飞行速度7.5马赫。该弹列装后，将使印度可以从孟加拉湾发射导弹覆盖欧亚非大部分陆地，大幅提高其海基核威慑能力。

Source: <https://mp.weixin.qq.com/s/KUGuz5MwRGigafGWsp2fQw>

Cruise Missiles

The BrahMos cruise missile, which shone in last year's India-Pakistan conflict, has passed combat testing, and India has built up considerable domestic production capacity — enough for its own use and large exports to the Philippines, Indonesia, Vietnam, and others. This missile's performance is at the international advanced level.

The BrahMos cruise missile is a new-generation cruise missile developed by the Russia-India joint venture BrahMos Aerospace. With supersonic speed and multi-trajectory characteristics, its

penetration, anti-jamming, and anti-interception capabilities are "unique" in the world — which is why the Indian and Russian media call it a "wonder" missile, the newest cruise missile. The BrahMos is 8.4 m long, 0.67 m in diameter, with a launch weight of 1,500 kg. Its propulsion system uses a solid-rocket booster plus an integral ramjet; its high-altitude trajectory range reaches 290 km. The body uses a dart-like aerodynamic layout, coated with radar-absorbing paint developed in India to enhance stealth. Its guidance combines inertial navigation and satellite navigation with terminal active radar homing. Since its first land test launch in June 2001, BrahMos Aerospace has spent over a decade improving it, gradually developing shipborne, land-based, air-launched, and submarine-launched variants, making it an internationally leading series of supersonic cruise missile weapon systems capable of striking surface and land targets. Judging from the performance reported in various sources, the series has the following characteristics: First, good stealth: the dart-like layout and the Indian radar-absorbing coating minimise the probability of detection by enemy air-defence radar. Second, high guidance accuracy: the missile combines inertial + active radar + satellite navigation, directly adopting the mature satellite navigation system used in Russia's KH-555 and KH-101 long-range strategic cruise missiles; in 2013, GPS-GLONASS combined positioning was added to the existing Doppler inertial platform, further improving long-range guidance accuracy. Third, high speed: the solid rocket booster plus liquid ramjet combination, with the main engine being a new small integral ramjet developed by India's HAL, flies at Mach 3 with a maximum range of 290 km. Fourth, strong manoeuvrability: the anti-ship variant has high and low trajectories, terminal sea-skimming flight, and terminal snake manoeuvres, descending to about 10 metres above the sea and flying in a serpentine path to evade enemy anti-missile interception; the land-attack variant has a large-angle diving attack capability suited to mountainous terrain. In short, the series is superior and has reached the world's leading level. Source: "A brief analysis of two domestically produced missile systems newly ordered by India's defence ministry" (WeChat article)



布拉莫斯是由俄印合资的布拉莫斯航空航天公司研制的新一代巡航导弹

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Air-Defence Missiles

Besides the S-400 bought from Russia, India has domestic air-defence systems such as the Akash missile. Within India, this missile is hyped almost as much as the BrahMos, mainly because it has produced real results in combat. The system has also been sold to Armenia, for use against Azerbaijan. Coincidentally, Azerbaijan has purchased China's HQ-9 systems and JF-17 fighter jets. If Armenia and Azerbaijan clash again, Indian equipment will once again face a comprehensive test against Chinese weapons. But the fact that India can sell this equipment abroad at least proves it is not entirely rubbish.

On key performance data: the Akash Mk1 missile weighs 720 kg in total, with a body diameter of 35 cm and a total length of 5.78 m, carrying a 55 kg fragmentation warhead detonated by radar fuze, with an effective lethal radius of about 10 metres. On range and altitude: its effective range is 25 km, maximum altitude 20 km; when the missile reaches 9,000 metres and the fuel burns out, its speed reaches 840 m/s. An Akash battalion comprises four missile batteries; the battalion's target detection and guidance is handled entirely by the Rohini radar, with a maximum detection range of 200 km and a very wide detection coverage angle. The Rajendra multifunction radar focuses on close-range operations, performing target search, acquisition, automatic tracking, and friend-or-foe identification within a radius of 60 km, while also handling missile guidance, building the technical support for every missile to hit its target precisely. Each Akash battery is equipped with one Rajendra radar operating in the 4.5–8 GHz band. This multifunction radar is outstanding: it can simultaneously guide two missiles

against the same target, handle up to four incoming targets, and continuously track 64 targets within 90 km. The scheduling of radars and launcher vehicles is coordinated by a battalion-level mobile command post, forming a complete operational loop. As of 2024, India has ordered a total of 32 Rajendra radars, steadily completing its air-defence system. At present, the Akash mainly handles point air-defence, mostly deployed around air bases. For example, Gwalior Air Force Station in Madhya Pradesh has two Akash battalions, fully replacing the previous S-125M systems. Compared with the S-125M, the Akash has better mobility and can deploy quickly without fixed positions. The battalions deployed at the base regularly move to field positions for realistic training; when tensions rise, they move forward as needed. During the India-Pakistan escalation of 2025, several Akash battalions were rushed to the border, shooting down several Pakistani reconnaissance drones — combat results validating the system's effectiveness. Source: <https://mp.weixin.qq.com/s/ORTli79Q3fmqIl4W-jHeKg>

Air-to-Air Missiles

India has developed an advanced domestic air-to-air missile with a range of over 100 km; by range alone it reaches the level of the PL-11. The PL-11 was test-fired in 2003, so this Indian missile may have reached China's level of 2003. India has the advantage of the latecomer; in the next decade we will very likely see rapid Indian development in the missile field.

According to India Today on 12 July, DRDO and the Indian Air Force jointly announced on 11 July 2025 the successful live-fire testing of the indigenous Astra beyond-visual-range air-to-air missile (BVRAAM). The test was conducted off the Odisha coast, using a Su-30 Mk-I fighter as the launch platform. During the tests, the team conducted two launch trials against high-speed unmanned aerial targets at different ranges, attack angles, and platform states. Both hit the intended targets, fully verifying the missile system's combat effectiveness. The missile uses an indigenously developed RF seeker with advanced guidance and navigation systems, with an effective range of over 100 km. The weapon system's development was led by several DRDO laboratories, together with HAL and more than 50 other public and private enterprises. Source: "India announces successful test launch of 'Astra' air-to-air missile with range over 100 km" (WeChat article)

Main Ground-Force Weapons

Tanks

India's Arjun tank has become a joke. But India's development of a light tank — quickly assembled by combining foreign engines and other key components — has produced a fairly decent highland combat platform. Like South Korea's weapon systems, this tank contains a great many foreign parts; but being able to integrate them shows that India's military industry has made considerable progress.

The Indian military's willingness to accept this tank in large numbers shows that, in its eyes, it is far better than the Arjun.



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Actually, the new "Zorawar" light tank is not the latest news. On 23 April 2021, India issued a request for information for its new-generation light tank. In August 2022, India's defence ministry issued a project brief for the new-generation light tank, officially named "AFV-ILT," and classified it under the "Make-I" category of Indian defence procurement. Under the "Make-I" category, 90% of project funding comes from government support, and most system components will be made by Indian domestic enterprises. On mobility: the Zorawar dropped the original design based on the K-9 tracked self-propelled howitzer's armoured chassis, adopting a new-generation tracked armoured vehicle chassis instead. Its power comes from a 750-hp diesel engine made by America's Cummins, giving a power-to-weight ratio of 30 hp/tonne, mated to a Renk transmission; maximum road speed is 65 km/h. The Zorawar also has amphibious capability: a detachable float module on each side of the hull and water-jet propulsion at the rear, with a wading depth of 1.5 m. With a combat weight of only 25 tonnes, it can be rapidly airlifted by the C-17 or Il-76MD heavy transports of the Indian Air Force. In firepower, the Zorawar mounts a two-person Cockerill 3105 turret made by Belgium's John Cockerill; its main armament is a 105HP 105 mm high-pressure rifled gun with a bustle autoloader; secondary weapons include a 7.62 mm coaxial machine gun, a remote weapon station with a 12.7 mm NSVT dual-purpose machine gun at the right rear of the turret, and a twin loitering-munition launcher on the

right side of the turret. The Zorawar also has day/night hunter-killer capability, using crosswind sensors and the PASEO modular advanced stabilised electro-optical sight developed by France's Safran — a multi-channel sight integrating day observation, third-generation thermal imager, laser designator, and laser rangefinder, with very advanced performance. Source: "India's new 'Zorawar' light tank makes public debut — how good is it?" (WeChat article)

In October 2025, during plateau trials, the Indian Army's Zorawar light tank successfully test-fired the "NLAW-style" (Spike LR2-type) NAG Mk II anti-tank missile — actually the "Vinaash"/"Dhruvastra" — the missile flew stably and hit the target in "top-attack" mode, verifying combat capability in complex mountainous terrain. The tank mounts a 105 mm tank gun and a launcher for the NAG Mk II missile (range 7–10 km), forming a composite fire system of "short-range direct fire + long-range precision strike." Maximum power 750–760 hp, unit power 30 hp/tonne, ensuring strong power output in low-oxygen, low-temperature environments; maximum speed 65–70 km/h, cross-country 35–40 km/h, wading depth 1.5 m. India's defence ministry has approved the purchase of 354 units; after the first 59 complete user trials, they are planned for induction in 2027, forming 7 light-tank regiments in total to strengthen the Indian Army's combat capability in the border highlands. Source: <https://mp.weixin.qq.com/s/3vAHhyDkHNhilh7mSGnaag>

Artillery

Besides the indigenous towed howitzer, India has developed a truck-mounted howitzer system. It is said to lag far behind China and other advanced countries, but the fact that it could be built at all shows India's military industry has a certain level.



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图 印度MGS155毫米卡车式自行榴弹炮

According to Army Recognition on 8 July 2025, the MGS 155 mm truck-mounted howitzer made by Bharat Forge has entered army trials, testing its performance across varied terrain and climates. The gun has a 52-calibre 155 mm barrel, designed for "shoot-and-scoot" artillery tactics. The MGS uses an 8×8 high-mobility chassis with a fully armoured blast-proof cab, shock-absorbing stabilisers, an onboard silent power system, and a modern fire-control system; rate of fire 6 rounds/min, range 45 km, able to displace in 85 seconds to evade enemy counter-battery fire. The gun is highly automated, with automatic gun calibration, positioning systems, and an integrated electronic control system, greatly reducing crew load and improving battlefield survivability. The MGS has completed internal trials at Balasore and Pokhran, firing over a hundred rounds and verifying reliability, and is now in competitive field trials. Its domestic content is as high as 85%, strengthening India's defence autonomy and reducing import dependence. Compared with traditional towed or self-propelled howitzers, it is more survivable against counter-battery fire, and fits the Indian Army's strategic shift toward a high-mobility, modular artillery force, able to cooperate with mechanised and armoured units. According to plans, the Indian Army needs 700–800 MGS truck-mounted howitzers, and a tender for 300 is currently under way. Source: "India tests MGS 155 mm truck-mounted howitzer" (WeChat article)

Besides the truck-mounted howitzer above, India recently test-fired a relatively advanced rocket system. Its range reaches 120 km — basically the level of China's PHL-03 rocket system; if production capacity is scaled up, it poses a certain threat.



US Defence Blog reported on 30 December that India completed the first flight test of its Pinaka long-range guided rocket (LRGR-120) at the Chandipur Integrated Test Range on the east coast on 29 December. Developed by a DRDO laboratory, the rocket was fired from an in-service Pinaka launcher, successfully verifying the 120-km maximum range, completing all preset in-flight manoeuvres, and hitting the target with "textbook precision," confirming the guidance performance and compatibility with the existing launch platform. Range instrumentation tracked the entire trajectory, verifying core parameters such as guidance and control at maximum range. The LRGR-120 is the longest-range guided variant of the Pinaka multiple-launch rocket family, and this test is an important result of the range's recent artillery and missile trials. Source: "India's long-range guided rocket completes first 120-km flight test" (WeChat article)

Attack Helicopters

To support operations against the imagined adversary in highland terrain, India developed a light attack helicopter based on the Dhruv. According to domestic military bloggers, this helicopter

matches the European Tiger, and is even stronger in highland conditions. India's LCH is slightly below China's WZ-10 but not by much — the LCH's engine is 900 kW, the WZ-10's is 1,000 kW, basically similar. But since the Dhruv has had several malfunctions and crashes, reliability may still lag Chinese equipment; that can only be made up gradually in actual service.



印度国产轻型武装直升机LCH

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印度LCH技术参数

最大起飞重量: 5.5吨

空载重量 3.243吨

主旋翼直径: 13.3 米

尾桨直径: 2.054米

机身长度: 13.94米

最大允许速度: 330 千米/小时

机高: 5.27米

最大巡航速度: 260千米/小时

实用升限: 6500米

航程: 700千米

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The LCH is a multi-role light attack helicopter developed and produced by HAL, capable of carrying out combat missions with modern weapons in complex climates and weather, and is held in high hope by the Indian military. However, as with many Indian weapons developments, performance long failed to meet expected targets and the schedule slipped badly; it was only formally inducted in 2022. To

minimise cost and airframe development time, the new LCH shares many parts and systems with the Dhruv. At present, the LCH's power system (including main and tail rotor systems and the integrated power system), launchers, electrical and hydraulic systems, and some avionics all come from the Dhruv. All the LCH's main systems have been tested on the Dhruv. The remaining systems, including weapon systems and some special equipment, will be tested on the armed Dhruv WSI helicopter. The Dhruv has successively used France's Turbomeca TM333 turboshaft engine, America's CTS-800-4N turboshaft, and the Aridden engine developed jointly by India and Turbomeca. After deliberation, India decided on the uprated Aridden engine jointly developed with Turbomeca, providing 900 kW, called "Shakti" (India calls it "Shivashakti"). The LCH uses a similar power system with two Aridden engines (each providing 900 kW), with a dual-redundant analogue flight control system. In overall technical performance, India's self-developed LCH is very close to Europe's Tiger attack helicopter. But compared with the Tiger, since the LCH is designed specifically for high-altitude operations, its performance at high altitude is stronger. The LCH can take off from airfields at 3,000 m, use its weapon systems at 5,000 m, and control UAVs at altitudes up to 6,500 m. For India, the LCH is indeed significant. On the one hand, massed attack helicopters give the Indian Army reliable supporting firepower, able to effectively suppress enemy armoured forces. On the other, the LCH can serve as a "mountain killer," supporting mountain operations and making up for the poor mobility and deployability of artillery and tanks. However, the LCH has only just entered service; whether it can deliver its expected combat power remains to be seen. Source: "Another star after the Dhruv: India's new LCH light attack helicopter" (WeChat article)

Drones

In the last India-Pakistan conflict, a number of private defence companies performed prominently; one of them, Zulu Defence Systems, is based in the tech hub Bangalore. Founded only a few years ago, it has already produced quite good drones. Besides this company, other startups are also working on drones. India's software and chip-design industries are developed, giving it an advantage in developing new technologies like drones.

Hovering fighters birthed in Bengaluru



Start-up Zulu Defence manufactures drones geared for deployment in warfare

By Haripriya Sureban

Updated - June 02, 2024 at 09:27 PM.



Zulu Defence co-founders (from left) Rajagopal Sethuraman, CFO; Nagendran Kandasamy, CEO; and Shashikumar Ramakrishna, CTO

India's defence procurement, long met through public sector undertakings apart from imports, is increasingly an area of focus for homegrown start-ups. Zulu Defence, a Bengaluru-based venture, manufactures drones geared for deployment in warfare.

Founded in September 2023 by Nagendran

Kandasamy, Shashikumar Ramakrishna and Rajagopal Sethuraman, the startup's area of focus is advanced air defence aerial systems for rapid deployment, surveillance and reconnaissance, precision strikes and

deployment, surveillance and reconnaissance, precision strikes, and



颠覆者：HOVERBEE^Q，这不是普通的四轴飞行器。它是一种可以放在手掌里的微型巡飞弹（自杀式无人机）。

这是一款紧凑型无人机，长15厘米，重260克，适用于室内/室外作战。它可以进入建筑物搜寻恐怖分子，投掷400克手榴弹，提供夜视功能，可进行2公里范围内的实时视频传输，即使在雾天/恶劣天气下也能出色地进行边境监视。

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DRAP（反装甲）：印度首款国产自杀式“坦克杀手”无人机。



除了微型无人机^o之外，他们还开发了 DRAP——一种更大的巡飞弹，配有5公斤重的弹头。它的目标是轻型车辆、掩体，甚至是坦克。

它就像“坦克杀手”，使步兵能够伏击机械化纵队，而无需使用重型反坦克导弹 (ATGM) 。

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VOLUME35^Q:

一种武器化的无人机平台，可以从空中发射 Igla-S 便携式防空导弹（肩扛式防空导弹），从而扩大防空范围，增强对敌方飞机/无人机的防御能力。



自力更生的印度优势，外国无人机通常成本高昂且存在数据安全风险。 Zulu提供了一种 100% 印度制造的解决方案，价格仅为美国同类产品（例如 Switchblade）的一小部分，并且专为高海拔边境地区设计。

Zulu Defence不仅仅是一家公司，它更是对对手的警告。印度天空中的眼睛比以往任何时候都更小巧、更智能、更致命。

Other companies' drones also have real capability; some have already seen combat.

鲁斯特姆-II：印度国产无人机



印度自主研发的鲁斯特姆-II无人机

由DRDO设计的Rustom-II（鲁斯特姆-II）无人机，代表了印度在国防技术上向自力更生迈进的努力。虽然该机尚未如一些外国制造的无人机那么多的行动，但它正在稳步证明它的价值。

主要用途：情报、侦察、监视（ISR）

技术参数：

- 有效载荷：350公斤，包括光电传感器和合成孔径雷达。
- 续航：12小时-理想的全天候监视。
- 航程：约180-200公里，未来升级将大幅提高。
- 装备：COMINT, ELINT、海上巡逻雷达、合成孔径雷达、中程电光、远程电光、隐形（机身3毫米凯夫拉尔和氯氟化碳）

实战运用：该机尚未进行实战运用，但经过广泛测试，达到25000英尺高度和10小时的续航，显示了该机未来用于边境侦察和作战的潜力。

采购成本：每台500-600万美元，由于国内制造，明显便宜。

维护：由于本土供应链，成本比进口无人机低20-30%。

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这款由印度初创公司IdeaForge开发的无人机，重量轻，便携，用途广泛。

主要用途：后勤、ISR

技术参数：

- 有效载荷：6公斤，通常配有监视摄像头。
- 续航：2小时，对于一个小型无人机来说很多。
- 航程：15公里，最大作业高度超过地面1公里，对于战术任务十分理想。
- 装备：可更换的有效载荷，25倍变焦昼间载荷和热成像夜间载荷，以及简单的基于路径点的导航。



一名印军士兵仔细观察进行武装测试的SWITCH无人机

实战运用：

2020年，在拉达克对峙中，印度军队订购了更多的该型无人机，为印军提供了在该地区恶劣地形下部队行动的实时图像。

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Air Force Equipment

Fighter Jets

India's Tejas fighter, like the Arjun tank, has become a joke among Chinese military fans. Having failed to develop a domestic engine, the Tejas uses an American engine. And military engines can easily be throttled for political reasons, so India's key task is to develop domestic engine manufacturing capability.

Since a fully self-developed engine is too difficult, India's current approach is to develop domestic production of Russian-type military engines, gradually mastering engine development capability and ultimately producing a domestic military engine. This development path seems rather similar to China's.

According to multiple media reports, GTRE, the gas turbine research institute under DRDO, has submitted a proposal to India's defence ministry to requisition two Su-30MKI fighters of the Indian Air Force, to be converted into flying testbeds. This may mean India is genuinely pushing its large-thrust military engine indigenisation programme. On 9 September 2024, India's defence ministry signed a production contract with HAL to produce 240 AL-31FP engines under Russian licence for the Su-30MKI. Under the contract, the engine's domestic content should reach about 54% in the early stage, rising to 63% by the end of the contract. Although the Indian government says the contract aims to "meet the Su-30MKI's continuing engine demand" — the products mainly replacing life-expired engines to keep the Su-30MKI flying — in practice, India can hardly acquire large-thrust engines from other channels or through independent development in the short term to meet the power needs of new-generation projects like AMCA. Therefore, steadily raising domestic content and gradually obtaining the core technology of the AL-31FP is very likely India's most realistic path to independent development of large-thrust military engines. Source: "India's research institutes requisition Su-30MKIs; large-thrust military engine development accelerates" (WeChat article)

Bombers

India has not yet developed a manned bomber, but it has produced a type of unmanned bomber. Domestic commentators find its shape peculiar and are unsure how to rate its combat power. But the aircraft is cheap, and its manufacturer plans to promote it in Africa.

In early May 2024, India's defence and aerospace company Flying Wedge exhibited its first unmanned bomber, the FWD-200B. On 3 September, the company announced the successful maiden flight of the FWD-200. Flying Wedge said the successful first flight of the first unmanned bomber represents a major advance for India's defence sector, realising the strategic goal of producing self-developed combat drones. The FWD-200B has a wingspan of 5 m and length of 3.5 m, a more compact design, 7

hours of endurance, a maximum take-off weight of 102 kg, and a payload of 30 kg — only 30% of the initial design. The FWD-200B cruises at 3,657 m, at a cruise speed of 152 km/h and a maximum speed of 250 km/h. Its minimum runway requirement is only 300 m, suitable for short runways. Compared with the FWD-200B display-board data of May, the real aircraft's performance has shrunk considerably: maximum take-off weight down from 498 kg to 100 kg, endurance down from 20 hours to 7. After the FWD-200B's first test flight, Flying Wedge said the drone's aerodynamic design, fuselage, power system, control system, and electronics were all developed at its production base in Bangalore's Electronics City — a great achievement. The company also said it is working with the Army Design Bureau leading the "Make in India" programme, hoping the drone will enter service with the Indian armed forces in the future. Source: "Equipment discussion: India's FWD-200B unmanned bomber makes successful first flight" (WeChat article)

Navy

Strategic Nuclear Submarines

India has built a domestic nuclear submarine — the first country after the five permanent members of the UN Security Council to develop one. Although some components come from France and elsewhere, the domestic content has gradually risen to 80%; and since France and India have excellent relations, it will most likely not be throttled in a potential regional conflict. In displacement, diving depth, and underwater speed, India's submarine is comparable to China's Type 092 — roughly China's level of the 1980s–90s. India is also developing a newer model, the S5. The S5 is expected to be delivered in the 2030s, with a displacement of 13,500 tonnes; besides the greatly increased displacement, there will be major improvements in propulsion, missile systems (expected to integrate the K-5), control systems, and more. By the 2030s, India will have a genuine sea-based nuclear deterrent against China; the probability of full-scale Sino-Indian war will fall further, and what will most likely break out is a war by proxy through Pakistan.



2016年,印度首艘国产核潜艇“歼敌者”号正式服役

Take the Arihant as an example: the Arihant-class submarine is 111.6 m long, 10 m wide, with a draft of 9.5 m, a surfaced displacement of about 5,500 tonnes and submerged about 6,000 tonnes, a crew of 95, and a maximum diving depth of 300 m. In development, the Arihant drew heavily on successful Soviet/Russian submarine design experience, such as the double-hull structure. Compared with single hulls, double-hull submarines have several advantages: with two layers of hull protection, they are far more collision-resistant and damage-resistant; double-hull boats have more ballast tanks, with reserve buoyancy of 25%–32% versus 10%–15% for single-hull boats, so they survive flooding better; in addition, double-hull boats have a lower construction threshold and can handle harsher sea states. In propulsion, the main power system consists of an 83 MW small nuclear reactor and turbines. The reactor is reportedly a copy of the Russian VM-4-1 design, built with Russian technical assistance. Maximum surfaced speed 12–15 knots, submerged maximum 25 knots, single shaft and single propeller with a 7-blade large-skew low-speed propeller, and a cruciform tail. In armament, the Arihant class has six 533 mm torpedo tubes in the bow, carrying 18 heavy torpedoes (6 in tubes, 12 stored in the torpedo room). In the missile bay in the mid-rear of the hull are four vertical launch tubes of 2.4 m inner diameter in a row, capable of launching the K-15 and K-4 submarine-launched ballistic missiles. Source: "The Arihant-class nuclear submarine India pins high hopes on — how good is it?" (WeChat article)

According to India's CHAKRA news website on 29 December 2025, India's fourth ballistic missile submarine, "S4*," has begun sea trials. The 7,000-tonne submarine departed from the Ship Building Centre (SBC) berth at Visakhapatnam port for sea trials last week. The S4* (four-star) is the last of the

Arihant-class ballistic missile submarines, capable of carrying eight K-4 submarine-launched ballistic missiles with nuclear warheads and a range of over 3,500 km. The S4*'s domestic content exceeds 80% — the highest of the four boats. The sea trials are expected to last a year, followed by formal commissioning in early 2027. India currently has four SSBNs, two in service and two in trials. The third, INS Aridhaman, has completed sea trials and is expected to be commissioned by the end of 2026. This strategic milestone marks the successful completion of a project begun in 1984. The project initially launched the Advanced Technology Vessel (ATV) programme, aiming to build three SSBNs. The S4* was ordered more than a decade ago as a transitional arrangement between the S4 and the larger S5 SSBN programme. Source: "India begins sea trials of fourth ballistic missile submarine and starts construction of the next-generation boat" (WeChat article)

India Begins Construction of Next-Generation S5 Nuclear Ballistic Missile Submarines

Published December 30, 2025 | By admin

SOURCE: AFI



According to a report published by chakranewz.com, India has already begun construction of the first two units of its next-generation S5 nuclear-powered ballistic missile submarines (SSBNs), marking a decisive step forward in the evolution of the country's sea-based nuclear deterrent. The S5 class represents a significant scale-up in capability and displacement, underscoring India's long-term commitment to maintaining a credible and survivable second-strike force.

With a reported displacement of around 13,500 tons, the S5 SSBNs will be nearly twice the size of the Arihant-class submarines currently forming the backbone of India's strategic undersea fleet. This substantial increase in size is expected to translate into greater endurance, improved stealth characteristics, higher payload capacity, and the ability to carry a larger number of longer-range submarine-launched ballistic missiles. Collectively, these attributes will allow the S5 class to undertake longer deterrence patrols while operating farther from home waters.

The report indicates that the first S5 submarine is expected to enter service in the early 2030s, following an extended period of construction, integration, and sea trials typical of strategic nuclear platforms. In total, India plans to induct four S5 SSBNs by the late 2030s, creating a more robust and continuous at-sea deterrence posture. Once operational, this fleet size would enable sustained patrol cycles, ensuring that at least one submarine remains on deterrent patrol at all times.

The initiation of construction on the S5 class also signals a transition from the learning-intensive phase of the Arihant programme to a more mature stage of indigenous nuclear submarine development. Lessons learned from earlier boats are expected to be incorporated into the S5 design, including improvements in reactor performance, acoustic quieting, command-and-control systems, and missile integration.

NOTE: Article cannot be reproduced without written permission.

Source: <https://idr.org/india-begins-construction-of-next-generation-s5-nuclear-ballistic-missile-submarines/>

Attack Submarines

India has imported technology from France and is building the Scorpène-class submarines locally. Brazil has also imported the Scorpène and is building it locally. India plans to build autonomous submarine development capability step by step through the P75I and P76 projects. The P75 is complete, the P75I has started, and the P76's status is unclear — probably not yet officially begun. India's P76 is not expected until the late 2030s or early 2040s. India's attack submarines would certainly not dare to cause trouble in the Pacific, but they pose many threats to Indian Ocean shipping lanes; any country deciding on war with India must consider this factor.

特点/方面	项目-75	项目-75I	项目-76
目的	建造六艘鲎鱼级常规柴电潜艇，技术转让来自法国（海军集团）。	在战略伙伴关系模式下建造六艘具有空中独立推进（AIP）和外国设计支持的先进常规潜艇。	建造 12 艘完全国产柴电潜艇，配备 AIP、先进的隐身和国内开发的武器系统。
土著内容	通过印度建筑技术转让适度（~40-50%）。	更高的本土含量（45%-60%）专注于新技术和本地制造。	非常高的本土含量（~90-95%），具有国产设计、武器控制和锂离子电池集成。
空气独立推进（AIP）	未配备（传统柴电）。	配备 AIP 燃料电池系统，可实现约两周的水下耐力。	配备 DRDO 的本土 AIP 系统，以增强耐用性和隐身性。
位移	约1,500-1,700吨（鲎类）。	比 Project-75 潜艇大，~3,000 吨。	约3,000吨水下排水量，下一代能力。
制造业	Mazagon Dock Shipbuilders（MDL）与法国合作。	获得外国 OEM 设计支持的印度造船厂（德国的蒂森克虏伯或西班牙的 Navantia）。	Larsen & Toubro（L&T）与 DRDO 合作，引领本土设计和建造。
运营时间表	2020 年代中期首次交付，2020 年代末至 2030 年代中期投入运营。	预计在 2020 年代末至 2030 年代中期之间交付（延迟）。	2027 年设计完成，2030 年开工，2037 年调试。
主要特点	隐身技术和现代鱼雷/导弹，但没有 AIP。	先进的传感器、隐身、巡航导弹能力、AIP 和 ISR 系统。	本土武器控制系统、锂离子电池、巡航导弹打击能力、隐身性强。
战略意义	现代印度常规潜艇部队的基础。	现代技术和更高的自力更生能力下一阶段。	向全面的本土水下海军能力迈进，重点是战略自主

Source: https://mp.weixin.qq.com/s/eFq_q8R-CLu-68bEGHzRtA

印度首艘国产攻击型常规潜艇下水

翔智科技信息 2015年4月9日 09:52

印度首艘“鲐鱼”级国产攻击型潜艇在孟买马扎兰造船厂下水，预计2016年9月服役。根据印法75项目计划，“鲐鱼”级潜艇由法国设计、印度建造，共6艘，本次下水是第一艘。预计“鲐鱼”级潜艇每九个月交付一艘，最后一艘将于2018年交付。

“鲐鱼”级潜艇排水量1700吨，共有6名士官和25名普通舰员，最多装载6枚SM 39“飞鱼”反舰导弹。

“鲐鱼”级潜艇将促进未来印度舰队的现代化。据悉，印度潜艇编队现有9艘俄罗斯“基洛”级潜艇（部分服役期已达30年），4艘德国209级潜艇，俄罗斯租借的“阿库拉-1”型核潜艇（印度名为“查克拉”号），以及1艘印度建造的“歼敌者”号弹道导弹核潜艇。

“歼敌者”号是印度建造的第一艘潜艇，该潜艇将于本月装载核导弹进行海试，预计2016年末服役。“歼敌者”级的二号舰已在2013年下水，预计今年交付。（中国船舶信息中心 陈晓）

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2025年9月11日 印度媒体报道：马扎冈船坞造船厂宣布，该船坞造船厂已开始与印度海军就潜艇项目P75(I)进行谈判。

今年8月23日，印度中央政府批准国防部与马扎冈船厂有限公司开始谈判，在德国的支持下，在印度建造六艘潜艇，该项目属于耗资7000亿卢比的“75India项目”。

印度国防部已选定马扎冈造船厂有限公司为中标者，德国蒂森克虏伯海洋系统公司为其技术合作伙伴。

该项目将增强印度海军实力，并为“印度制造”计划做出贡献。此外，拉森与特布罗公司还将参与建造两艘**核攻击潜艇**。

蒂森克虏伯海洋系统公司首席执行官表示，此次合作将创建一个能够在未来几年内为全球潜艇市场提供服务的生态系统。

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<https://mp.weixin.qq.com/s/VoWqJlOXNYjO8UI2cZ-BmA>

Destroyers

India's Kolkata-class destroyers are mediocre; facing the air and naval forces of a great power, their survivability is questionable. India has now proposed the P18 destroyer, benchmarked against the

great powers; Chinese military fans doubt India can complete the project. But with India's improving basic industry and, under great-power competition, easier access to European technology, it is not impossible that combining European, Israeli, and other technologies could produce a serviceable destroyer. If it really succeeds, the Indian Navy's presence in the Indian Ocean will greatly increase.



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Indian Navy Seeks AoN for 11,000-Tonne P-18 Destroyers to Dominate Indian Ocean in 2040s

 Raghav Patel · Yesterday at 11:39 AM · Featured News

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According to Indian Navy data, the P-18-class destroyer is expected to have a full-load displacement of about 11,000 tonnes — the largest destroyer in India's history, far exceeding the in-service P-15A Kolkata class (7,500 tonnes) and P-15B Visakhapatnam class (8,000 tonnes). Its main radar will be a new-generation dual-band large active phased-array radar using gallium-nitride technology, the indigenously developed "LRMFR system." It will use an integrated mast incorporating SIGINT and COMINT suites and an advanced electronic-warfare system. The hull design emphasises stealth. For propulsion, it will use a British advanced hybrid integrated electric propulsion system. In armament, it will carry 96–128 vertical launch cells for Israel's Barak-8ER long-range air-defence missiles (later replaced by India's indigenous "Kush" long-range air-defence missile), VL-SRSAM missiles, and the Indo-Russian BrahMos-2 hypersonic missile (all under development; the BrahMos-2 will be based on Russia's Zircon missile), and is planned to mount 50–100 kW high-energy directed-energy weapons, with positions reserved for experimental electromagnetic railguns or hypersonic projectiles. Construction will be assigned to India's two main shipyards, Mazagon Dock and Garden Reach, with overall design by the Indian Navy's warship design bureau. But this does not mean the ship will be entirely "Made in India": the plan mentions selecting a foreign partner to assist with the systems, especially the design and integration of the combat system; the Indian Navy puts the domestic content at about 70%–75%. Earlier, India Defence claimed "this is a modern large warship with stealth

capability, expected to become the world's most powerful surface combatant in the future"; but from the specifications one can see it is basically a scaled-up Type 052D. Source: <https://mp.weixin.qq.com/s/p34Fx-QKWEi2Ip6zgy47wA>

Aircraft Carriers

India's domestic aircraft carrier is mediocre — adequate against Pakistan, but it could easily be wiped out in a conflict with China. India has now proposed building a domestic nuclear-powered carrier with electromagnetic catapults. Since carrier propulsion and electromagnetic catapult systems are far harder than submarines, and only a handful of countries possess the technology, importing the technology would be difficult for India; it depends on self-development, so the probability of completing the project on schedule is basically zero — more likely the 2040s. However, India does have a certain nuclear foundation, its carrier aircraft are supported by France and Russia, and the electromagnetic catapult will have to be self-developed — so it is not entirely impossible.

According to Belgium's Army Recognition on 8 September, India recently issued its "2025 Technology Perspective & Capability Roadmap," proposing the development of a nuclear-powered aircraft carrier. Possibly to be named INS Vishal and built by Cochin Shipyard, it would be about 300 m long, displace 65,000–77,000 tonnes, and reach 30 knots; it would mount two electromagnetic catapults (40-tonne launch capability), an automatic landing system, a Fresnel lens optical landing aid, arresting gear, and combat-management software dedicated to carrier aircraft command and control; it could carry 40 fixed-wing carrier aircraft and 15 helicopters. At the same time, India plans a combination of short-term purchase and long-term self-development to form a more diverse and technologically advanced carrier air wing for the future nuclear carrier, including 26 Rafale-M fighters from France, twin-engine carrier aircraft to be developed and delivered in the 2030s, and unmanned combat aircraft. Source: "Indian Navy plans to develop a nuclear-powered aircraft carrier" (WeChat article)

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